

PHISHING EMAIL DETECTION & AWARENESS SYSTEM

Future Interns – Cyber Security Internship

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1. INTRODUCTION

Phishing is one of the most common cyber-attacks used by cybercriminals to steal sensitive information such as usernames, passwords, banking details, and personal information. Attackers often impersonate trusted organizations and send fraudulent emails to trick users into revealing confidential information.

The purpose of this project is to analyze phishing emails, identify suspicious indicators, classify email threats, and spread awareness regarding email security practices.

2. OBJECTIVE

The objectives of this project are:

- Identify phishing emails and suspicious messages.
 - Analyze characteristics commonly used in phishing attacks.
 - Classify emails as Safe, Suspicious, or Phishing.
 - Create awareness regarding email security.
 - Recommend preventive measures against phishing attacks.
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3. TOOLS USED

- Google Search
 - Browser
 - Email Samples
 - Microsoft Word
 - GitHub
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4. EMAIL ANALYSIS

Subject: Urgent: Verify Your Bank Account

Classification: PHISHING

Analysis:

This email attempts to create urgency by informing the user that their bank account has been suspended. The email contains a suspicious link and requests immediate action. Such tactics are commonly used by attackers to steal banking credentials.

Risk Level: HIGH

Screenshot 1 – Phishing Email (Fake Bank Alert)

From: security-update@secure-bankverify.com ↗

Subject: Urgent: Verify Your Bank Account

Body:

Dear Customer,

Your bank account has been temporarily suspended due to unusual activity.

Please verify your account immediately by clicking the link below:

<http://secure-bankverify-login.com>

Failure to verify within 24 hours will result in account closure.

Regards,

Bank Security Team

Subject: Password Reset Required

Classification: PHISHING

Analysis:

The email claims that the user's password is about to expire and provides a suspicious link for password reset. The sender domain appears fake and does not belong to the legitimate company.

Risk Level: HIGH

Screenshot 2 – Phishing Email (Password Reset)

From: support@microsoft-security.com ↗

Subject: Password Reset Required

Body:

Dear User,

Your Microsoft account password will expire today.

Click below to reset your password:

<http://microsoft-password-reset-login.com>

Thank you,

Support Team

Subject: Invoice Attached

Classification: SUSPICIOUS

Analysis:

This email contains an unexpected invoice attachment from an unknown sender. Such emails are often used to distribute malware through malicious attachments.

Risk Level: MEDIUM

Screenshot 3 – Suspicious Email

From: invoice.department@gmail.com ↗

Subject: Invoice Attached

Body:

Hello,

Please review the attached invoice and make payment immediately.

Attachment: Invoice.zip

Regards,

Accounts Team

Email Sample 4

Subject: Interview Schedule

Classification: SAFE

Analysis:

The email is sent from a verified company domain and contains professional communication. No suspicious links or malicious attachments are present.

Risk Level: LOW

Screenshot 4 – Safe Email

From: hr@company.com ↗

Subject: Interview Schedule

Body:

Dear Candidate,

Your interview is scheduled on Monday at 10:00 AM.

Meeting Link:

<https://meet.google.com/abc-defg-hij>

Regards,

HR Team

Email Sample 5

Subject: Project Meeting Update

Classification: SAFE

Analysis:

This email contains normal workplace communication from a known sender. The content is relevant and does not attempt to collect sensitive information.

Risk Level: LOW

Screenshot 5 – Safe Email

From: manager@company.com ↗

Subject: Project Meeting Update

Body:

Hello Team,

The project review meeting has been moved to Friday at 3 PM.

Regards,

Project Manager

5. COMMON PHISHING INDICATORS

The following signs help identify phishing emails:

1. Fake URLs

Attackers often create websites that closely resemble legitimate websites.

2. Urgent Language

Messages create panic and force users to act quickly.

3. Spelling and Grammar Errors

Many phishing emails contain poor grammar and spelling mistakes.

4. Unknown Senders

Emails from unfamiliar sources should be treated cautiously.

5. Suspicious Attachments

Unexpected attachments may contain malware.

6. Requests for Sensitive Information

Legitimate organizations rarely request passwords or banking details through email.

6. RISK ASSESSMENT

Threat	Risk Level
Credential Theft	High
Malware Infection	High
Identity Theft	High
Financial Fraud	High
Data Leakage	Medium

The most significant risk associated with phishing attacks is credential theft, which may result in unauthorized access to personal or organizational accounts.

7. PREVENTION GUIDELINES

To protect against phishing attacks:

- Verify sender email addresses carefully.
 - Avoid clicking unknown or suspicious links.
 - Inspect website URLs before entering credentials.
 - Use strong and unique passwords.
 - Enable Two-Factor Authentication (2FA).
 - Keep software and operating systems updated.
 - Install trusted antivirus software.
 - Avoid downloading unexpected attachments.
 - Report suspicious emails immediately.
 - Participate in cyber security awareness training.
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8. AWARENESS RECOMMENDATIONS

Organizations should implement the following measures:

- Conduct regular cyber security awareness sessions.
- Deploy email filtering and anti-phishing solutions.
- Encourage employees to report suspicious emails.
- Establish incident reporting procedures.
- Monitor email communications for malicious activity.

- Perform regular security assessments.

These measures help reduce the risk of successful phishing attacks.

9. CONCLUSION

Phishing attacks continue to be one of the most common and dangerous cyber threats affecting individuals and organizations worldwide. Through this project, different email samples were analyzed and classified based on their characteristics and risk levels.

The analysis demonstrates the importance of identifying suspicious indicators such as fake links, urgent language, unknown senders, and malicious attachments. By following cyber security best practices and maintaining awareness, users can significantly reduce the risk of becoming victims of phishing attacks.

Continuous education, vigilance, and proper security controls are essential for maintaining a secure digital environment.

10. REFERENCES

1. OWASP Security Guidelines
2. Cyber Security Awareness Resources
3. Google Security Center
4. Public Phishing Email Samples
5. Email Security Best Practices